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Design Space Exploration of Verifiable Credential Schemas using Partial Graph Modeling

Scientific Students' Association Report

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Kivonat

Az ellenőrizhető tanúsítványok (Verifiable Credentials) szabványosított használatának megjelenésével az információ ellenőrzők egy tervezési tér felderítési (Design Space Exploration) problémával szembesülnek: milyen típusú ellenőrizhető állításhalmazokat kérjenek személyektől és szervezetektől az ellenőrzési igényeik kielégítésére, úgy, hogy egyrészt maximalizálniuk kell az állítások ellenőrzésének megbízhatóságát, másrészt minimalizálniuk kell a megosztott adatok mennyiségét, és azok identitással való korrelációjának kockázatát.

Az egymással összefüggő állításhalmazok tervezése során meghozott döntések befolyásolhatják az ellenőrizhető tanúsítványokra támaszkodó döntési folyamatok hitelességi és adatvédelmi jellemzőit, különösen ahogy nő az értelmezett állítások halmaza. Az algoritmikus modellezés segíthet a tervezési kompromisszumok azonosításában, a tervek rangsorolásában, és a rendelkezésre álló szabványok közötti választásban. A megjelent szakirodalomban a tervezési döntések következményeit kis állításhalmazokra vizsgálják, nagyobb és összetettebb állításhalmazokra még nem. Emellett még nem létezik algoritmikus megközelítés az egymással összefüggő ellenőrizhető állításhalmazok tervezésére. Specifikusan az adatvédelem kontextusában a következő kérdés merül fel: Hogyan maximalizálhatjuk az összefüggő ellenőrizhető tanúsítványok kifejező erejét, miközben fenntartjuk a birtokosuk identitása feletti önrendelkezését a megosztott adatok mennyiségének és a korreláció kockázatának minimalizálásával?

Ezt a tervezési problémát parciális gráfmodellezéssel és -generálással közelítem meg. A parciális gráfmodellezés alkalmas egy ellenőrzési forgatókönyv információs gráf alapú követelményeinek kifejezésére és az úgynevezett ellenőrizhető prezentációk sablonjainak generálására. Ez lehetővé teszi a bevált tervezési tér felderítési technikák alkalmazását, ideértve a lehetséges tervek sorrendezését és értékelését párhuzamos szempontok alapján.

Ebben a dolgozatban egy szisztematikus módszert javaslok az egymással összefüggő ellenőrizhető tanúsítványok tervezéséhez. Ismertetem az ilyen állításhalmazok lehetséges tervezési szempontjait, különös tekintettel az adatvédelemre, és részletezem megközelítésem a lehetséges tervek generálására információs gráf alapú követelményekből parciális gráfmodellezés segítségével. A tervezési döntések és az e területen kialakulóban lévő szabványok összetettségének kezelése érdekében a generált modellek a Model Driven Architecture megközelítést követik, amely lehetővé teszi a tervek precízebb elemzését és biztosítja a szabványoknak való megfelelést.

Az általam javasolt módszer hiánypótló az összetett ellenőrizhető tanúsítványok algoritmikus tervezése területén. Ez a megközelítés lehetővé teszi az ellenőrizhető tanúsítványok tervezői számára, hogy optimalizálják az állításhalmazok kifejező erejét, miközben elkerülik a korreláció kockázatát és a szükségtelen adatközlést. Javaslatom alapot teremt a kifinomultabb és adatvédelem-tudatosabb ellenőrizhető tanúsítvány kezelő rendszerekhez, javítva a digitális identitásmegoldásokba vetett bizalmat.

Abstract

With the advent of the use of standardized Verifiable Credentials, information verifiers begin to face a Design Space Exploration problem: what kinds and potential sets of verifiable claims to request from persons and organizations to meet their verification needs, especially so that on the one hand, they have to maximize the trustworthiness of claim verification, but on the other hand, also observe the principle of data minimization and the correlation risk minimization needs of credential holders.

The design of interrelated claim sets may influence the credibility and privacy characteristics of decision processes which rely on Verifiable Credentials as sources of truth, especially as the number of managed claims increases. Algorithmic modeling approaches help in identifying trade-offs, ranking designs based on relevant metrics, and in choosing from the available standards. Surveying existing research, the implications of design decisions are being explored for small claim sets, but the analysis remains incomplete for larger and more complex sets of claims. Also, there are yet to be any algorithmic approaches for designing interconnected Verifiable Credential schemas. Specifically, in the context of privacy characteristics, the following question arises: How do we maximize the expressive power of a related set of verifiable credentials while maintaining their holder’s control over their claims by minimizing the amount of data shared and the risk of correlation?

My proposal is to approach this design problem through partial graph modelling and generation. Partial graph modeling can be amenable to express the potentially diverse information graph requirements of a verification scenario and generate admissible templates for so-called verifiable presentation graphs. This facilitates the application of established Design Space Exploration approaches, including the enumeration of options integrated with multi-aspect evaluation and rank ordering.

In this paper, I provide a systematic method for designing interrelated Verifiable Credential schemas. I explore the possible design considerations of such claim sets, with a focus on privacy, and detail my approach for generating possible designs from information graph requirements using partial graph modeling. To tackle the complexity of both design decisions and emerging standards in this field, the generated models follow the Model Driven Architecture approach, which enables more precise reasoning over the designs and ensures compliance with existing standards.

My proposed approach addresses the current gap in algorithmic methods for creating complex Verifiable Credential schemas. It allows designers to optimize the expressiveness of claim sets while avoiding the risk of correlation and unnecessary data disclosure. This approach lays the foundation for more sophisticated and privacy-conscious credential management systems, improving trust in digital identity solutions.

Chapter 1

Introduction

In recent years, Verifiable Credentials (VCs) [1, 2] have stirred the interest of not just researchers but also legislators and corporations in the realm of digital identity management, including the European Union [3, 4, 5, 6]. They are a crucial part of a novel and proliferating identity model called Self-Sovereign Identity (SSI), which aims to offer its users more interoperability, privacy, and control over their identity on the Internet. VCs enable individuals and organizations to exchange trusted digital claims in a privacy-preserving and interoperable manner. However, as these credentials become increasingly integrated into complex business processes and regulatory frameworks, their design poses significant challenges that extend beyond simple data representation.

A critical challenge emerges as organizations begin adopting VCs: how to structure interrelated sets of claims into credential schemas that serve multiple verification scenarios while maintaining privacy, security, and usability. This challenge is particularly acute in regulated sectors such as finance, healthcare, and government services, where credentials must satisfy complex compliance requirements while protecting sensitive personal data. For instance, a financial services ecosystem might need to verify claims ranging from basic identity information to detailed financial history, employment records, and existing credit obligations - all while minimizing data disclosure and correlation risks.

To address this challenge, I propose an approach that combines three complementary methodologies:

1. **Partial Graph Modeling** for the systematic generation of credential schema designs
2. **Model Driven Architecture** principles for ensuring interoperability and requirement satisfaction

3. **Design Space Exploration** techniques for evaluating and selecting optimal designs

My aim is to design a comprehensive framework for systematically designing sets of interrelated Verifiable Credential schemas that satisfy complex requirements while maintaining privacy and security characteristics. To this end, I use Partial Graph Modeling (PGM) [7] to generate diverse VC architectures, as the VC data model can be expressed as **subject-property-value** relationships, and thus VCs can be modeled as Information Graphs. To help systematically ensure interoperability, and the satisfaction of complex requirements the generated designs follow the MDA approach. These generated designs form a design space, which can be explored with generic or use-case specific metrics. The approach enables designers to explore and evaluate different schema architectures systematically, moving beyond manual design approaches that become intractable as complexity grows.

My contributions include:

- A systematic method for generating diverse credential schema designs using PGM
- A MDA-based approach ensuring interoperability between different VC formats
- An iterative framework for evaluating generated designs against diverse requirements
- A proof-of-concept implementation demonstrating the feasibility of automated credential schema generation

In Chapter 2, after a brief introduction to Verifiable Credentials, the role they fill in SSI, and the design challenges they present, I detail how VCs can be modeled as knowledge graphs and explain key concepts from DSE and MDA that form the foundation of the approach.

In Chapter 3, I present the emerging design problem of Verifiable Credential Schema Ecosystems, and examine both generic and domain-specific requirements that influence their design. The chapter argues for approaching this challenge through systematic methods, particularly the combination of Partial Graph Modeling with Model Driven Architecture principles to enable comprehensive Design Space Exploration.

In Chapter 4, I detail the approach for generating Verifiable Credential Schemas from Knowledge Graphs, introduce the architecture of model generation and explain how it implements MDA's abstraction layers. The chapter describes the partial meta-model and discusses how the generated models can be evaluated with established DSE techniques with a proposed process.

Finally, Chapter 5 concludes my work by summarizing the contributions and outlining directions for future work, including extending the meta-model, implementing comprehensive evaluation metrics, and validating the approach with industry use-cases.

Chapter 2

Background

In this chapter, I introduce Verifiable Credentials (VCs) in the wider context of Self-Sovereign Identities (SSI). I explain how they can be modeled as knowledge graphs, and how graph-based requirements aid the design of VCs through an illustrative example. Then I briefly introduce DSE, with some common approaches and metrics with which designs are evaluated. Finally, I summarize the relevant concepts from the MDA approach. In the following chapters, I use these concepts to construct my approach for systematically designing interrelated sets Verifiable Credential Schemas.

2.1 VCs in an (increasingly) SSI world

Verifiable Credentials (VCs) are a novel family of digital claims. They are conceptually part of the SSI, which according to Brian Behlendorf, GM for Blockchain, Healthcare, and Identity at the Linux Foundation, is “the most crucial fix for today’s broken Internet.” [8]

Self-Sovereign Identity (SSI) - also sometimes called *Decentralized Identity* - is a collection of ideologies, design paradigms, protocol frameworks, cryptographic principles, and, recently, software solutions that aim to provide an alternative to the existing identity management paradigms. Contrasting with today’s centralized and federated models, it is a decentralized, peer-to-peer identity architecture based on the principle that each peer has total control over the identifiers they own or are subject to. Within this model, VCs are the standardized and digitally verifiable form of the trusted claims and information which the parties share about each other. By using VCs, parties have a cryptographic framework for distributing claims in a peer-to-peer and privacy-preserving way. Many “flavors” of VCs also support the

use of novel Zero-knowledge Proof (ZKP) techniques, which when integrated into applications, minimizes the unnecessary information shared by the users. They have been adopted by the European Union in the European Digital Identity Wallet [3, 6] and in the EU Dataspaces initiative [5, 9].

In this section, I discuss Self-Sovereign Identity with an emphasis on VCs, as laid out by Drummond Reed and Alex Preukschat [8] as well as the Trust over IP Foundation’s model [10]. My aim is to outline how VCs work and argue that their design is non-trivial.

2.1.1 Concepts of Self-sovereign Identity

Digital identity is the sum of all digital information that exists connected to a person, such as their name, address, birth date, education and work history, health and insurance data, and financial information. It describes who they are in every aspect of their digital life.[11] SSI is an approach to digital identity which aims to give people and organizations full control over their data, through structural and technological innovation.

SSI allows us to model digital identity just like we are used to in the non-digital world based on paper documents and cards. Instead of documents made of paper and plastic stored in a wallet, data is shared through digital credentials stored in wallet applications on the user’s phone. Importantly, these digital credentials can be securely shared in a peer-to-peer manner, and their authenticity can be reliably verified. [8, 12]

The basic lifecycle of these credentials is usually illustrated with the so called *Trust Triangle* [8, 11, 10], seen on Figure 2.1, which describes the different roles and their relationships regarding data sharing.

Issuer, Holder and Verifier [13]

- **Issuers** - Parties who “issue” identity-related data to people or organizations (“Holders”) in the form of digital credentials. They are the original data sources of a digital identity ecosystem. For example, a government issues digital passports to citizens or a university issues digital diplomas to graduates.
- **Holders** - Individuals or organizations who receive digital credentials that contain data about themselves from various sources (“Issuers”). By aggregating and storing such credentials in digital wallets, Holders can build holistic digital identities that are under their control and can easily be shared with third parties (“Verifiers”).

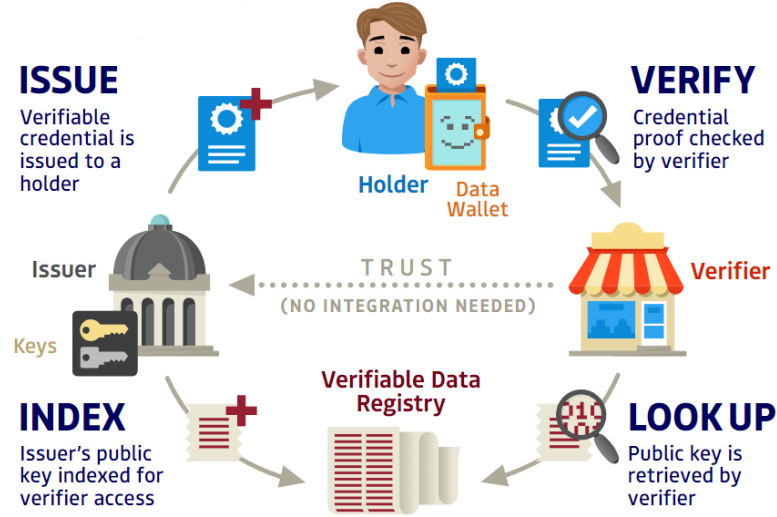


Figure 2.1: Trust Triangle from the Trust over IP Foundation

- **Verifiers** - Parties who rely on data to provide products and services can reliably verify and process data that has been provided by others (“Holders”). Verifiers, also called “Relying Parties”, are usually organizations or individuals in their professional capacity.

To model the technological and functional requirements of these relationships, the Trust over IP Foundation defined a protocol stack [10] (Figure 2.2), through which I will present the technological building blocks of SSI based systems. Their goal is to present SSI as a possible *trust layer* over the internet.

The lower two levels of the stack focus on meeting the technical requirements of digital trust with *Verifiable Data Registries (VDRs)* serving as repositories of cryptographic material digital trust can be based on, and *Trust Spanning Protocols* - communication protocols enabling secure and private communication. The top two layers focus on meeting human requirements with Verifiable Credential Exchange protocols, which can be integrated into specific but standardized software ecosystems. Through the ToIP protocol one can establish trusted, secure, and private peer-to-peer connections, issue, exchange, and verify digital credentials, and store public credential data on Verifiable Data registries, which may use decentralized or distributed record-keeping technologies.

In the next few sections, I cover the relevant technological building blocks of the ToIP protocol stack[10], and explain how interoperability can be maintained between the different VC formats, as well as what challenges the design of VCs poses. Even though the building blocks of the ToIP stack [10] *are* related, they are often implemented independently, and may also exist as independent services. Their discussion should follow similar compartmentalization.

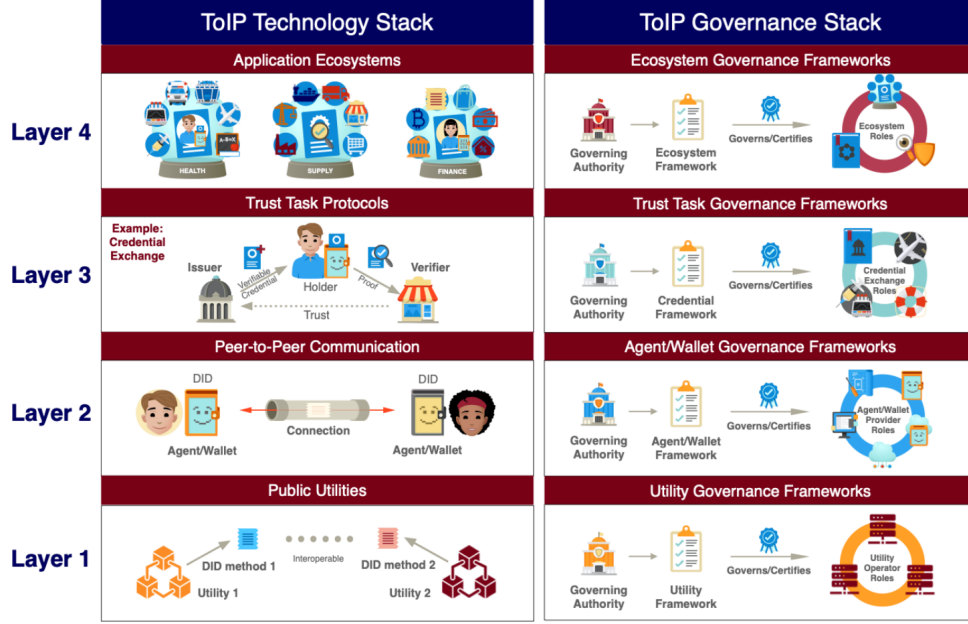


Figure 2.2: Trust over IP model

2.1.1.1 Decentralized Identifiers

Decentralized Identifiers (DIDs)[14] give us the power to verify information, like credentials, anywhere, anytime, through the establishment of a novel form of public key infrastructure.

DIDs are unique identifiers (URIs) which are standardized by the W3C [14]. They can refer to any subject - from a person to an organization, to a thing or basically anything else for that matter. They associate this subject with a DID document (DIDDoc). A DIDDoc contains cryptographic information with which the controller of the DID may prove ownership of the DID, or ensure the authenticity of the claims made by them in a digital credential.

A DID is reachable through a DID method, which is a technical specification of how the identifier can be resolved. DID methods may be implemented over various types of verifiable data registries, such as the Domain Name System, existing Public Key Infrastructure (e.g., X.509 or JWS), general-purpose blockchains [14, 15], purpose-built blockchains [16], or even using a peer-to-peer model (pairwise and n-wise DIDs).

2.1.1.2 Existing VC formats

The W3C defines a Verifiable Credential (VC) as “... a tamper-evident credential that has authorship that can be cryptographically verified.” While there are

many competing flavors and models of VCs, the core of the abstract data model is established by the W3C Verifiable Credentials Data Model [1].

VCS encode claims about their Subject made by their Issuers, cryptographically linked to both their Issuer and their Subject. These claims can then be verified by presenting them to a Verifier - meaning the claims are communicated through a secure channel, with the instructions and resources (e.g. signatures and public keys, possibly from PKI or a DIDDoc) to verify their authenticity, and whether they have been revoked. Verifying a VC does not directly imply the truth of the claims encoded in it, only their authenticity and currency. Rather, after authenticity and currency are established, a verifier may evaluate the claims based on their own policy and trust in the Issuer.

VCS are held within digital wallet applications, which are digital solutions designed to store VCS securely, and with the ability to present these credentials using various *presentation protocols* (See Section 2.1.1.3).

There are multiple competing encoding formats of VCS, (See Table 2.1) which offer different privacy characteristics and are supported by different *presentation protocols*. A key privacy characteristic is whether they support some kind of ZKP protocol, and if so, then what kind. A **Zero-knowledge Proof** is a cryptographic scheme where a Prover is able to convince a Verifier that a statement is true, without providing any more information than that single bit (that is, that the statement is true rather than false). In the context of VCS, this usually means the following two types of schemes:

Selective Disclosure enables a holder to choose which pieces of information contained in a VC will be revealed to a verifier, rather than being forced to reveal all the data present in a VC, through the use of cryptographic protocols, such as BBS [17]. For example, Alice can share only her age to verify being old enough to purchase products offered in an e-commerce shop, without revealing other personal information present in her Verifiable ID document used for verification. [18]

Range Proofs is a ZKP that demonstrates a numeric value in a credential that falls within a specified range (e.g., between a minimum and maximum), without revealing the actual value. Now, Alice shares only *the fact* that she is old enough, without revealing her actual age.

As encoding formats can be derived from the abstract data model, a modeled claim set can be represented in either of them. If a designer has an abstract model for a set of interrelated VCS, they can derive the format specific representations, and choose the best one for their needs. This can be heuristic or even systematic, as

some characteristics of the formats - e.g. whether they support Selective Disclosure - can be modelled.

Table 2.1: Comparison of Verifiable Credential formats

Format	Owner	Published	Enc.	SD	Pros
JWT	IETF	2015-03	JSON	No	Widely adopted and supported, simple to implement and use
CWT	IETF	2018-05	CBOR	No	Adopted and supported, but not so much as JWT Smaller binary payload than JWT
W3C VC JSON-LD	W3C	2024-10	JSON	No	Semantically expressive, extensible with linked data
SD-JWT VC	IETF	2024-09	JSON	Yes	Supports SD while being compatible with existing JWT infra
AnonCreds	LF DT	2024-03	JSON	Yes	Strong privacy features, such as SD and range proofs
mDocs (mDL)	ISO/IEC	2021-09	CBOR	No	Standardized for mobile driving licenses
JSON Web Proof	IETF	2024-09	JSON	No	Extends JWT with proof mechanisms

To enforce a specific structure on a given claim collection, the W3C defines *Data Verification Schemas*, which are used to establish that the structure and contents of a VC conforms to a published schema. Mostly they enumerate what claims the VC must contain. A *JsonSchema*, for example, lists what keys are in the VC, and what is their data type. Many ecosystems call these schemas **Verifiable Credential Schema (VCS)**, which is the terminology used in this paper.

2.1.1.3 Presentation protocols

A final technical detail of the ToIP model [10] is the layer through which VCs can be communicated, **presentation protocols**. They are sometimes called *Credential Exchange Protocols*, *Trust Spanning Protocols*, or *Transfer Envelopes*. These protocols (See Table 2.2) enable the secure and private communication of claims on- and sometimes offline, while offering a diverse set of complementary functions which aid developers in the design of processes and applications. They influence the choice of

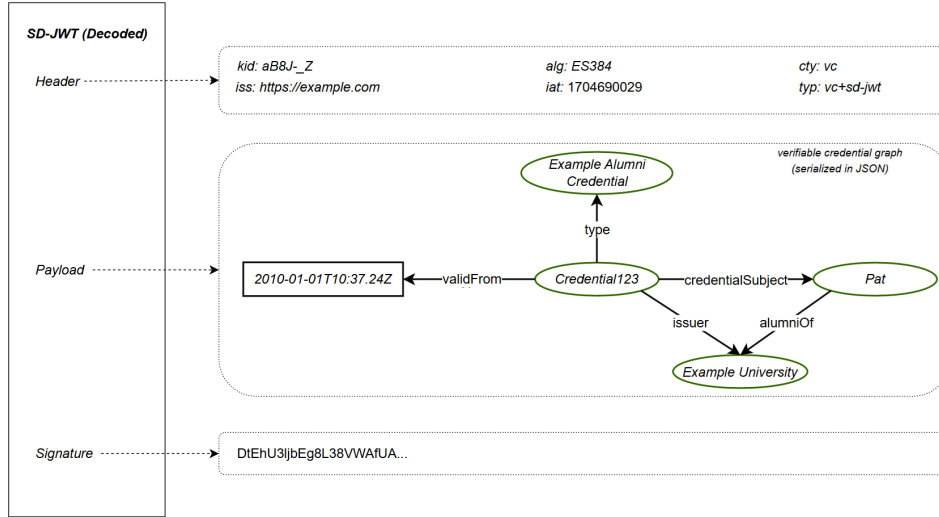


Figure 2.3: Information graph associated with a basic verifiable credential; source: W3C Verifiable Credential Data Model v2

VC format as each supports different ones. They are important in the context of regulatory compliance, as they are the central pillar of security in a digital process involving VCs.

Table 2.2: Comparison of Verifiable Credential exchange protocols

Protocol name	Owner	EUDIW	ARF	Compliance	Proximity
OpenID for Verifiable Credentials	OIDF	Explicit	for remote flows		Bluetooth
Mobile Driver's License(mDL)	Auth0	Explicit	for proximity flows		NFC
DIDComm	DIF	Possible			BLE
Aries	LF	Unlikely			BLE
	DT				
Credential Handler API 1.0 (CHAPI)	W3C	Likely			-

2.1.2 Credentials as Knowledge Graphs

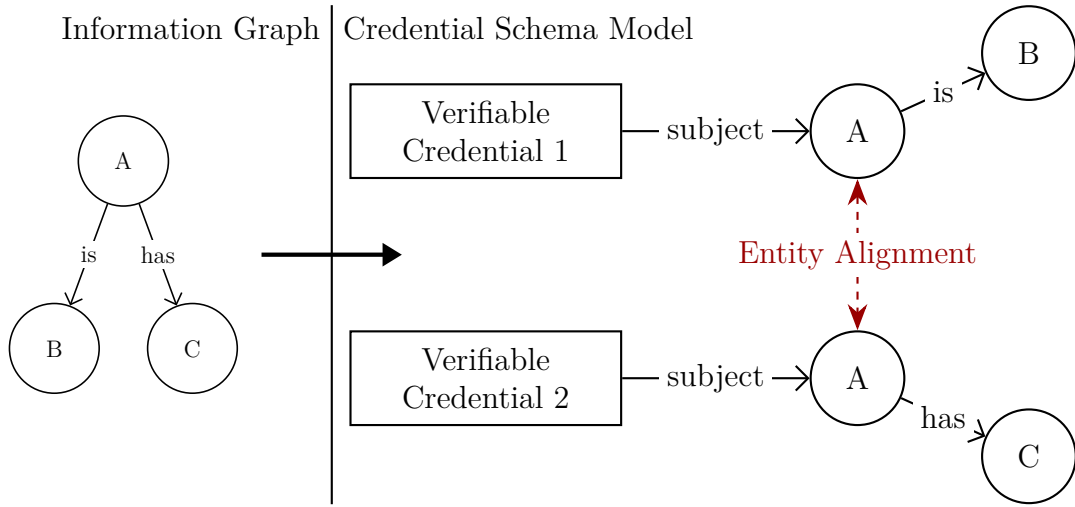
Claims in VCs can be expressed as **subject-property-value** relationships, e.g. "Tom"-**"has"**-**"blue eyes"**. This means that VCs can be modelled as Knowledge Graphs (KGs) or Information Graphs (IGs).

As Verifiable Credential Schemas (VCSs) make establishing the structure of common claim sets easier; their design influences the privacy characteristics of a Verifiable Credential ecosystem. How the claims are divided up between credentials plays a crucial role in how expressive a set of related VCSs is and how precisely a Holder

can control what they disclose when they present a VC. This, and many other requirements of VCSs can be modeled as graph-based requirements.

An important concept in this case is *Entity Alignment*, which is the process of identifying nodes that represent the same entity across multiple KGs. In our case, if two Verifiable Credential Schemas (VCSs) contain claims about the same subject, there needs to be a process that enables us to identify them and thus enables us to reason over the joint IG, as shown in Figure 2.4.

Figure 2.4: Decomposition of an information graph into verifiable credential schemas and illustrating entity alignment



Reasoning over Information Graphs constructed from VCSs allows us to make inferences and evaluate policies over subjects in a privacy-preserving way. Although many small examples have been demonstrated in existing research (See in Chapter 3), a systematic design approach has yet to be proposed for VC schema sets over which a policy may reason.

I define a set of interrelated VCSs, which are designed together, have a related set of requirements, influence each other's design characteristics, and can form a connected Information Graph as a **Verifiable Credential Schema Ecosystem (VCSE)**.

2.1.2.1 A Motivating Example

To demonstrate how such graph-based requirements come about, I present a hypothetical VCSE in which I can highlight the challenges that lie in their design:

A national government along with a consortium of financial institutions is developing a comprehensive financial services ecosystem. This system needs to assess individuals for various financial products and services, including credit scoring, mortgage

applications, investment opportunities, insurance policies, but also regular KYC procedures, simple wire transfers, and payment with credit cards. To highlight one *complex* scenario in this ecosystem, I explain what kinds of credentials a client must present to get access to government supported mortgage for newly constructed homes based on the Hungarian “CSOK” government program [19].

Credentials presented for a government supported mortgage

- Debtor and Co-debtor:
 - National ID - to prove their identity and that they are Hungarian
 - Proof of residence - To prove that they live in Hungary
 - Proof that they are not under foreclosure or bankrupt - BAR lista
 - Account at the bank
- Marriage and children:
 - Marriage certificate - To prove that they are married
 - Birth certificate of the child
- Employment:
 - Proof of Social Security payment status
 - Financial statements for three months of pay - So that they have enough income to pay the mortgage
 - Proof of employment for at least three months of *continous* and open-ended employment
- Constructed home:
 - Proof of ownership of the used plot
 - Construction permit

The presented claims come from multiple issuers, and have widely differing characteristics. For them to be trustworthy and reusable, the designers of the ecosystem must carefully consider how the claims are divided between the credentials in this scenario, along with the claims in the previously mentioned simpler ones. Here are a few important aspects of the claims contained within the credentials:

Design aspect of claims contained within VCs

Persistence and update frequency

Long-term or static Never or rarely changes *e.g. birth date, education*

Periodical Evaluated on a regular refresh cycle, or valid for a certain period
e.g. financial statements; concert ticket

Event-triggered Changes based on specific events *e.g. address change*

Highly Dynamic Information Real-Time Data, continuous updates
e.g. account balance

Sensitivity/Risk associated with unwanted disclosure (financial; identity; reputational)

High Risk *e.g. Identity Card Number, Bank Details*

Medium Risk *e.g. Address History, Employment*

Low Risk *e.g. Public Certifications, Skills*

Privacy/Correlation risk of claims

Correlation Risk Could enable unwanted correlation

e.g. Location + Purchase History

Inference Risk Could enable unwanted inference

e.g. Medical History + Insurance Claims

Aggregation Risk Reveal more when combined

e.g. Social Media + Financial Activity

Data Relationship Complexity

Independent Claims Can be verified standalone *e.g. Basic Identity*

Hierarchical Claims Parent-child relationships, claims depending on each other

e.g. Organization -> Department -> Role

Network Claims Complex interconnected relationships

Revocation propagation Patterns

Atomic Revocation Individual claim can be revoked *e.g. Single Account Status*

Group Revocation Related claims that must be revoked together

Cascading Revocation Revocation triggers other revocations

e.g. Professional License -> Related Certifications

Requirements about how claims fit into different credentials along these design aspects can be represented as graph patterns - e.g. static and dynamic claims cannot be part of the same credential means that they cannot be connected in the sub-graph modeling the credential. Because of this, the generation of graph models which satisfy these patterns enables the designers to easily create compliant designs.

In the next few sections, I detail the methods and approaches I used in the creation of the proposed design flow. Specifically, how do we generate these models, how can we represent them in a way so that these patterns can be made explicit, and how designers may choose from the generated models.

2.1.3 Partial graph modeling

Partial Graph Modeling (PGM) is a modeling technique used to specify and generate graphs that satisfy certain structural constraints and requirements. Instead of manually defining, or deterministically generating graphs, we can:

- Model known and unknown relationships between nodes

- Define wanted and unwanted patterns
- Specify how inferences are made in the generation process
- Generate multiple, consistent graphs

This approach enables the generation of *multiple* valid models that satisfy complex, graph-based requirements, if given a sufficient starting partial model.

During PGM, we start with a partial model which consists of:

- A **metamodel** or *domain* specifying nodes and edge types and how they can be connected;
- **Predicates** which declaratively denote patterns in a given metamodel;
- **Error predicates** which denote unwanted patterns, suitable for effectively shrinking the design space;
- **Instance models** which are specific model parts that are required to be present in the generated model;
- and the **model scope** which denotes the desired size range of the entire generated graph, with the option defining size ranges of specific model elements.

This defines what is certainly *known* and what is yet *unknown* in the model. From this, a modeling tool can generate multiple compliant models. This is done by representing the model using 4-valued logic - *true*, *false*, *unknown*, *error*, and applying valid transformation steps until there are no unknowns, and there are no errors. This process is called *graph refinement*. When the process is finished, a single consistent, valid model.[20]

2.1.3.1 Used technology: Refinery

Refinery is a partial graph modeling and generation tool with which we can generate provably diverse graphs from a starting partial model using graph refinement with efficient, state-of-the-art algorithms. Its main benefits are:

Domain Specific Language (DSL) for defining partial models. An easy to learn modeling language based on first order logic, similar to Prolog in this aspect.

Low computational complexity because it stores not just the model but all truth values in memory, enabling faster execution with a larger memory footprint.

Large expressive power based on 4-value predicate logic, which allows the easy declarative definition of arbitrarily complex partial models.

Graph refinement meaning in each solution step, the solver takes an *unknown* element to be resolved by decision rules.

Propagation rules - language construct for defining specific refinement steps - can be thought of as rules in 4 value logic.

Generation of provably diverse graphs is beneficial, as in DSE, it is imperative to choose from a diverse set of possible designs.

With Refinery, we have a tool for modeling knowledge graphs, and generating multiple designs which satisfy complex graph based requirements, which can be serve as a basis for DSE (See Chapter 3).

2.1.4 Design Space Exploration

Design Space Exploration (DSE) is a systematic approach to analyzing and evaluating different possible designs that satisfy a given set of requirements. It involves exploring a *design space* to find solutions that best meet specified objectives. [21, 22, 23] This *design space* represents all possible combinations of design decisions and their resulting solutions.

The primary goal of DSE is to help designers understand and navigate complex trade-offs between different design objectives. This includes:

- Identifying feasible design alternatives
- Understanding relationships between design decisions
- Evaluating designs against predefined criteria
- Finding optimal, near-optimal, or most satisfying solutions

A key challenge in DSE is that design problems often have competing objectives - improving one aspect may worsen another. For example, in chip design, increasing processing speed typically increases power consumption. The goal is often not to find a single “best” solution, but rather to understand the trade-offs and identify a set of promising alternatives.

In the previously stated example, it is trivial to identify competing and non-competing design objectives. An example of the former would be between VCSE complexity and VCS complexity. The more complex individual VCSs are allowed to be, the less complex the relationship between them in the ecosystem representing the starting claim set.

2.1.4.1 Possible approaches for exploring the design space

The process of DSE can be categorized by how the design space is explored. These approaches, although may contain similar steps, have distinct relationships with the design space.

Approaches for Design Space Exploration based on exploration type

Exhaustive exploration

- *Systematically examines all possible combinations of design decisions*
- Guarantees finding optimal solutions, if they exist
- Limited to small design spaces due to combinatorial explosion
- Example: Testing all possible CPU cache configurations up to a certain size

Guided exploration

- *Uses domain knowledge and heuristics to focus on promising areas*
- More efficient but may miss optimal solutions
- Often employs sampling strategies or expert rules
- E.g. Starting with known good architectures and exploring variations

Evolutionary exploration

- *Starts with initial designs and evolves them through iterations*
- Uses genetic algorithms or other metaheuristics
- Particularly useful for large, complex design spaces
- Example: Evolving neural network architectures for specific tasks

My approach fits into the *Guided exploration* category, because after generating some models, designers evaluate on domain specific metrics. Model driven approaches have shown to be feasible in similar scenarios, as shown by the work of Abel Hegedüs, Akos Horvath, and Daniel Varró [24], which partly inspired my work.

2.1.4.2 Evaluation techniques

Design alternatives encountered during each of the exploration approaches can be evaluated. The results of the evaluation of these design points - alone or together - inform which design is chosen in the end, and may even be a key part of some approaches (e.g. genetic algorithms discard models that are worse than the previous iteration, and further mutate those who are better in some metric)[21].

These evaluations may happen either analytically on the model, by measuring an implementation, or by simulation.

I list some evaluation methods, which are relevant to my use-case.

DSE evaluation techniques

- **Ranking based on single-objective functions** If we have some sortable function of the characteristics of the designs which we want to optimize, then we can simply rank the designs by this function, and find the most fitting one.

- **Pareto Analysis of multi-objective functions** Sometimes we have multiple optimization objectives in a DSE process. Since the objectives are often in conflict, there cannot be a single optimal solution that simultaneously optimizes all objectives. In this case, we can analyze the *Pareto efficiency* of the designs, and choose from the *Pareto front* the optimal designs form. [21]
- **Constraint verification** Another metric is whether the design meets some formalized requirement. If the design doesn't pass the filter, it is scrapped.
- **Manual expert driven** An approach for small design spaces where designers manually analyze the model with a domain expert, who can help by providing heuristics, and implicit requirements, which when made explicit, can be incorporated into an automated design process.

2.1.5 Model Driven Architecture

Model Driven Architecture (MDA) is an approach to software design and development that provides a set of guidelines for structuring specifications expressed as models. Introduced by the Object Management Group (OMG) in 2001, MDA emphasizes the separation of business or application logic from the underlying technological platforms. The aim is to derive value and automate the path from high-level stakeholder focused models to executable information systems. [25, 26, 27]

Abstraction layers A core principle of MDA is that “It is useful to identify particular “layers” of an architecture with respect to its level of abstraction.” [27] We will borrow the following abstractions:

Relevant abstraction layers in MDA

- Computation Independent Model (CIM):** model of actual, *real* things in a specific domain, not representations of those things in an information system. Describes “business” concepts without thought for architecture. Also known as the domain model or business model.
- Platform Independent Model (PIM)** models components of a system and their interactions, without the specific used resources in mind, describes system functionality.
- Platform Specific Model (PSM)** Combines PIM specifications with details about how the system uses a specific platform or technology. It is derived from the PIM by specifying the technology that is being modeled. The goal of PIMs is for developers to be able to directly generate implementation code from the final model.

Transformations A featured value proposition of MDA is the ability to relate and automate the generation of such models. In this aspect, MDA serves to separate

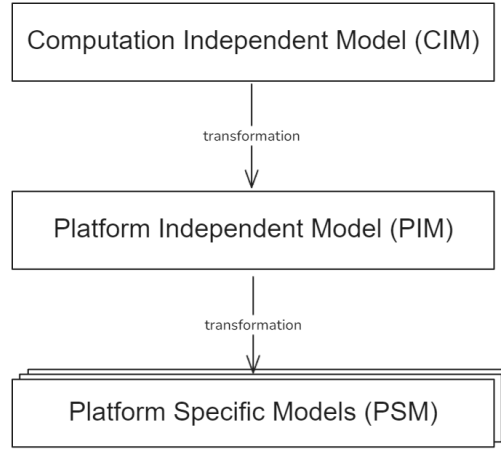


Figure 2.5: MDA Layers

the concerns between those business system requirements and the technology that implements them. This is achieved through defining *transformations* between the models. [27]

Transformation is the process of applying a pattern to produce one representation from another, or to cross levels of abstraction or architectural layers. The two abstract patterns used in this work are the following:

Relevant transformations in MDA

CIM to PIM Transformation A pattern mapping business requirements to system functionality, preserving business rules and constraints.

PIM to PSM Transformation Adds platform specific technical details and parameters, maintaining functional equivalence across platforms.

Later, in Section 3.1.4, I argue that the MDA approach provides a solid foundation for systematic VC schema design, enabling the creation of interoperable, maintainable, and privacy-preserving credential ecosystems. By separating concerns at different levels of abstraction, it allows designers to focus on the essential aspects of credential design while ensuring the satisfaction of complex requirements and the consistent implementation across different platforms and formats.

Chapter 3

Design Space Exploration of Verifiable Credentials

In this chapter, I argue for approaching the design of VCSs with systematic methods. I discuss the different characteristics that *generic* and *domain specific* VCSEs may have, and review some existing designs along this spectrum. I present requirements which are amenable to a PGM process. I argue for a Model Driven Architecture (MDA) based approach, in order to systematically satisfy requirements and enable cross-format interoperability. Finally, I discuss the benefits of systematic model generation and the resulting DSE process.

3.1 An Emerging design problem

As organizations begin adopting Verifiable Credentials (VCs) for complex business processes, they face increasingly challenging design decisions about how to structure the claims they want to represent into interrelated Verifiable Credential Schemas (VCSs), which need to serve multiple verification scenarios with different privacy requirements. These design decisions impact the privacy, security, and usability characteristics of the resulting Verifiable Credential Schema Ecosystems (VCSEs).

Consider the previous example of a financial ecosystem (See Section 2.1.2). The system needs to verify claims ranging from basic identity information to detailed financial history, employment records, and existing credit obligations. These claims come from multiple issuers - government agencies, employers, other financial institutions - and vary significantly in their structure, sensitivity, revocation pattern, and interdependence.

The design decisions made when structuring these claims into credential schemas have far-reaching implications - from privacy and verifiability to usability. There are generic and domain specific characteristics and interdependencies between the claims, that often form evaluable patterns. As the number of claims and their interdependencies grow, this *design space* becomes increasingly difficult for designers to navigate manually.

The challenge is further complicated by technical constraints and emerging standards. Different credential formats offer varying privacy features - from simple JSON-LD [1] to sophisticated ZKP-supporting AnonCreds [28]. Designers must consider not just the logical grouping of claims, but also how these groups map to the technical capabilities of different formats, especially when interoperability between formats is required. As VCSEs grow to encompass more issuers, subjects, and verifiers, these design decisions have an increasingly significant impact on the overall system’s privacy, efficiency, and trustworthiness.

This emerging complexity in VCS design creates a need for systematic approaches for exploring and evaluating different schema architectures. Without formal methods to analyze design alternatives, credentials risk being either too coarsely grouped (compromising privacy through unnecessary disclosure and revocation) or too finely separated (increasing complexity and correlation risks). The following sections present my approach to this challenge, demonstrating how DSE techniques can be applied to VCS design to systematically evaluate these trade-offs and generate “optimal” schema organizations for specific use cases, while also providing interoperability between different VC formats.

3.1.1 Application specific usage of VCs

As VCs can represent arbitrary claims, there are some already explored use-cases where they offer a benefit to existing data transfer systems. As shown by their adoption by the European Union [6, 3, 5, 9], they are suitable for natural persons and legal entities, and to be used in sector specific use cases with complex domain requirements (*data markets*), as well as the generic representation of widely understood and used claims (*electronic identification*).

I want to highlight the latter spectrum, particularly that the generic claim sets have categorically different stakeholder and requirement sets from sector specific ones. They need to be adopted by nation states and intergovernmental organizations, and must satisfy the needs of potentially billions of entities using them rather than the fewer, but still numerous organizations within an industry sector. Of course, there

are relationships between generic and sector specific design choices (*e.g. KYC processes should accept national eIDs*), with generic choices influencing sector specific ones.

I argue that the design process I lay out in this work is suitable for both, but practically, because the adoption process in business sectors is more hierarchical than democratic, and because the requirement sets - even though more complex - are more cleanly defined, the benefits of my process are more pronounced in sector specific VCSEs.

Verifiable Credential Ecosystem categories with examples

Generic and widely used

- Electronic Identification (eID)
- Cross-border certifications
- Diplomas and education credentials

Sector Specific

- Supply chains
- Telecommunications
- Data collection with IoT devices, i.e. Smart meters
e.g. gas, electricity, weather
- Financial services

3.1.2 Relevant state of the art

Surveying existing publications, there is yet to be a systematic process for designing Verifiable Credential Schemas or Verifiable Credential Schema Ecosystems. Still, there is inspiration to be gained from published large-scale, manual design processes.

From the generic end of the previously defined spectrum, there is the *European Union Digital Identity Wallet [6], which is a framework for member states to implement SSI-like wallet applications and identity systems in the EU. One key design consideration they take is to designate a special set of claims - the “PID” or “Personal Identification Data” credential - which are under much higher scrutiny from a privacy and truthfulness perspective, because the unwanted disclosure or correlation of the claim set poses a paramount risk for the individual. Because of these higher guarantees, they can be even be used in contexts which require such high level of assurance, such as the tax-collection or medicine. The key takeaway is that defining risk for unwanted disclosure or correlation for a claim or set of claims, and making design decisions based on that risk enables higher levels of assurance and the mitigation of those risks.

On the other end of the spectrum, I found a few proposed Verifiable Credential Schema Ecosystems for specific industries. First, there has been a proposal for a highly scalable operating model for digitization and trust of cross border trade based on verifiable credentials [29], which was showcased by Amazon [30] as a valuable use-case for governments to adopt. They demonstrate that by using VCs, and carefully considering their design, while incorporating the needs of specific Issuers and Verifiers, they can improve efficiency, and mitigate risks associated with inauthentic goods, while maintaining existing, paper-based business processes.

Second, Garzon et al. [31] prototype a 5G Service-Based Architecture that utilizes VCs instead of traditional X.509 certificates and OAuth2.0 access tokens to authenticate and authorize network functions. In their prototype, they propose a hierarchical model of trust delegation using VCs between Issuers, which decentralizes trust relations when it comes to issuing AuthN and AuthZ VCs. My key takeaways from their research is that even a technically specific field - Telecommunications - can benefit from the adoption of VCs, and that the design of a VCSE is not only influenced by the simple design aspects a specific claim poses in and of itself, but also the structures and hierarchies they form.

3.1.3 Requirements in interrelated Verifiable Credential Schema Ecosystems

As I explained in Section 2.1.2, there are many design aspects associated with claims within Verifiable Credentials (VCs). How claims are divided between VCs poses diverse risks to Issuers, Holders, and Verifiers alike, as thoroughly discussed by the W3C in their standard [1]. Some of these risks can be mitigated at design time by defining requirements which make the harmful or exploitable state unrepresentable. In this section, I show how some of these unwanted scenarios can be avoided by considering how claims are distributed across credentials. As VCs can be modelled as Knowledge Graphs (KGs), these distributions can be represented as graph patterns.

Bundling dependent claims Consider the following example borrowed from the W3C [1]: “A university is issuing two VCs to an individual. Each VC contains two properties that, when combined, indicate the person’s”role” in a specific “department.” For instance, one VC pair might designate “Staff Member” in the “Department of Computing,” while another could signify “Post Graduate Student” in the “Department of Economics.” Atomizing these VCs results in the university issuing four separate credentials to the individual. Each credential contains a single des-

ignation: “Staff Member”, “Post Graduate Student”, “Department of Computing”, or “Department of Economics”. The Holder might then present the “Staff Member” and “Department of Economics” VCs to a Verifier, which, together, would comprise a *false* claim.”

This scenario can be avoided if the pattern of putting a dependent claim in a standalone credential is disallowed. We can easily reify this pattern using a predicate in the Partial Graph Model. Specifically, when we have a hierarchical dependence of claims, we can define the pattern that a credential for a child claim always contains the claim directly depended upon. Doing this upwards the whole dependence tree ensures that when presenting the whole tree or the relevant sub-tree, the Holder cannot represent the false claim laid out in this scenario.

Static-dynamic claims Another example to consider is when the revocation pattern of the claims within a credential are different. Given a passport credential which contains a person’s name - constant throughout their life - and the claim they can travel abroad - revoked periodically. When such a credential is put to use, the claims which have a longer “lifespan” are unnecessarily revoked, and could even result in the Holder being unable to present a claim that could be considered valid not for the unnecessary revocation. In these cases, where the revocation patterns of the claims mismatch, then they should be put into *separate* credentials. In a sense, this is the opposite of the previous example, as this requires that a credential *does not* contain two claims with different revocation patterns.

3.1.4 The case for an MDA Approach

The design of VCSEs for the purpose of DSE presents unique challenges that align naturally with the principles and benefits of MDA. The choice of MDA emerged from the following three considerations regarding the design of VCSEs:

MDA specific considerations for VCSE design

- The need for systematic requirement satisfaction
- The imperative for interoperability between different VC formats
- The complexity of managing multiple stakeholder needs across different abstraction levels

3.1.4.1 Systematically ensuring the satisfaction of requirements

My approach for DSE of VCSs utilizes the systematic generation of diverse models. Because of this, it is imperative to use an approach which systematically ensures the satisfaction of several, potentially complex requirements. MDA offers several key advantages in this regard:

Benefits of MDA in systematically meeting requirements

Separation of Concerns: MDA enables the modeling of requirements of the abstraction layers independently, while also systematically ensuring these requirements are met throughout the entire process.

- Business requirements can be modeled independently of technical implementation details
- Privacy requirements can be enforced at the conceptual level before technical decisions are made
- Security constraints can be verified across different levels of abstraction

Traceable Transformations: The *transformations* defined in an MDA process provide traceability between the model elements across abstraction levels. This traceability is useful for maintaining consistency and managing changes in complex credential ecosystems. Specifically:

- Each transformation from CIM to PIM to PSM maintains clear linkages
- Requirements can be traced from business needs to technical implementation
- Changes in requirements can be systematically propagated through the model chain

Validation at Multiple Levels: MDA's layered approach enables validation of different aspects of the credential ecosystem at the most appropriate level of abstraction:

- Business rules can be validated at the CIM level
- Structural constraints can be checked at the PIM level
- Technical compliance can be verified at the PSM level

3.1.4.2 Enabling interoperability between carrying formats

Another compelling benefit of adopting MDA in VCSE design is its natural fit for handling multiple VC formats. MDA's platform-independent modeling approach allows designers to define the abstract model of a VCSE without being constrained by format-specific requirements. By establishing patterns for abstract requirements at the PIM level, designers can focus on the essential characteristics of the abstract

model without getting entangled in the technical details of specific VC formats. This abstraction enables changes to be made at a conceptual level, ensuring that modifications to credential structure or behavior can be implemented without concern for format-specific constraints.

Clear transformation rules enable the generation of implementations in various formats such as JSON-LD [1], SD-JWT [32], and AnonCreds [28]. Also, this approach provides natural extensibility - support for new VC formats can be added by defining new transformation rules without altering the abstract VC model.

MDA ensures consistent semantics across different VC formats. This semantic consistency is particularly valuable in complex credential ecosystems where different participants may prefer or require different formats while needing to maintain interoperability.

3.1.5 The expected benefits of Design Space Exploration

The expected benefits of applying DSE to Verifiable Credential Schema Ecosystems (VCSEs) design span multiple dimensions, from immediate technical advantages to longer-term ecosystem benefits.

At the most fundamental level, DSE enables systematic exploration of schema partitioning options while ensuring compliance with privacy and security requirements. By generating and evaluating multiple valid designs, it allows designers to discover non-obvious solutions that might be missed through manual design approaches. This is valuable when dealing with complex credential ecosystems where the interaction between different credentials can create subtle privacy implications.

Rather than relying on intuition alone, DSE enables the evaluation of different designs based on specific metrics. For example, designers can measure the granularity of credential schemas - how coarsely or finely they are grouped, or the correlation risks between different credential combinations. This quantitative approach enables the identification Pareto-optimal solutions where neither privacy nor security nor complexity can be improved without degrading each other.

Also, the systematic nature of DSE provides strong guarantees about requirement satisfaction. Specifically, that all generated designs provably meet the specified constraints, and also that conflicts between requirements are identified early in the design process. This is especially valuable in regulated environments where credential schemas must satisfy complex compliance requirements while maintaining usability.

Perhaps most importantly, DSE provides designers with insights into the design options and their implications. As credentials are generated and evaluated, patterns emerge that help inform future design decisions. This learning aspect is valuable because:

Insights gained during an iterative DSE process

- It helps identify implicit constraints that weren't obvious in the initial requirements
- It reveals unexpected but valid design alternatives
- It builds institutional knowledge about useful credential schema patterns

In the context of evolving VCSEs, these benefits compound over time as designers can leverage insights from previous iterations. DSE naturally supports an iterative design process where:

Steps of an iterative DSE process

1. Initial designs are generated and evaluated
2. Requirements are refined based on evaluation results
3. New design constraints are added to address discovered issues
4. The process repeats with a refined design space

This iterative capability is valuable for VCSEs that need to evolve as requirements change or novel use cases emerge.

Through these complementary benefits, DSE shifts the process of credential schema design from relying primarily on expert intuition to a systematic approach, while still preserving the flexibility needed to address novel requirements and use cases. This combination of systematic rigor with design flexibility is valuable in the rapidly evolving field of VCs, where both technical capabilities and real-world applications are rapidly expanding and maturing.

Chapter 4

Approach for generating VC Schemas from KGs

To provide a systematic method for designing Verifiable Credential Schema Ecosystems (VCSEs) which contain claims from a predefined Knowledge Graph (KG), i.e. which credential should contain which claim from a starting Information Graph (IG), I propose a **Design Space Exploration (DSE)** [22] process using Partial Graph Modeling (PGM) [7], which produces templates for platform-specific VCSs, over which VCs can be issued, from which - when presented by their Holder - the required IG can be partially or fully re-constructed and reasoned over by the Verifier. These designs follow the MDA approach, for the reasons detailed in Section 3.1.4.

The three prongs of my approach (DSE, PGM, and MDA) complement each other in interesting and important ways, as I detail in this chapter. At the outset, I would like to highlight PGM allows for the generation of multiple diverse designs while also serving as an impeccable basis for creating models which satisfy complex requirements within and across the MDA abstraction layers. Also, the use of MDA following models in a DSE process enables designers to compose metrics for *each abstract step* of the modeling process along with the final model.

To generate the models, I use Refinery [20]¹, a partial graph refinement and graph generation tool, which can generate diverse graphs based on a ruleset written in its own DSL. It requires a partial model as its input, as discussed in Section 2.1.3. Based on this input, it is able to generate diverse compliant models. Because the models are generated using PGM, a designer may define only a part of the model, and still get a complete model in the end. Specifically, the model can be refined with use-case-specific elements, predicates, and instance model. Using PGM with a solver offers another benefit: If the partial model is *unsatisfiable* - meaning it already

¹Online version of the tool is available at <https://refinery.services/>

represents an invalid model, from which no complete models can be generated - the designer can be sure that they made an error *before* generating any models.

The design process goes as follows:

Flow of the design process

1. The designer defines a partial graph model extending an abstract meta-model, which contains the graph-based requirements they want the models in the searched design space to satisfy.
2. Multiple diverse models are generated using Refinery. These models make up the design sub-space that is explored.
3. The models are evaluated by DSE techniques, such as the ones mentioned in Section 2.1.4.2
4. A chosen model can be implemented based on the carrying format specific representation of the VCSs in PSMs

Through this process, a designer can generate diverse verification requirement setups for their specific use case, which satisfy complex requirements, and create an interoperable, potentially provably privacy-preserving Verifiable Credential Schema Ecosystem.

4.1 Architecture of model generation

As the starting point of the partial model, I define a generic metamodel and predicate set, which generates models that reflect the core logic of MDA. This metamodel defines how the generated models are structured, and what constraints they must satisfy. The generated model can be divided into three subgraphs analogous to the steps of MDA. Figure 4.1 provides an overview of the architecture and the structure of the generated graphs. As stated before (See Section 3.1.4), the sub-graphs model the abstraction layers in the design process. The specific abstraction layers of the design process, and the transformations between them are detailed in the next few sections.

4.1.1 Abstraction layers following the MDA approach

The metamodel from which the designs are chosen in the end consists of three abstraction layers, each modeling a part of an MDA design flow. First there is the Knowledge Graph (KG) we want to include in the Verifiable Credential Schema Ecosystem (VCSE). From this, an abstract Verifiable Credential Schema (VCS)

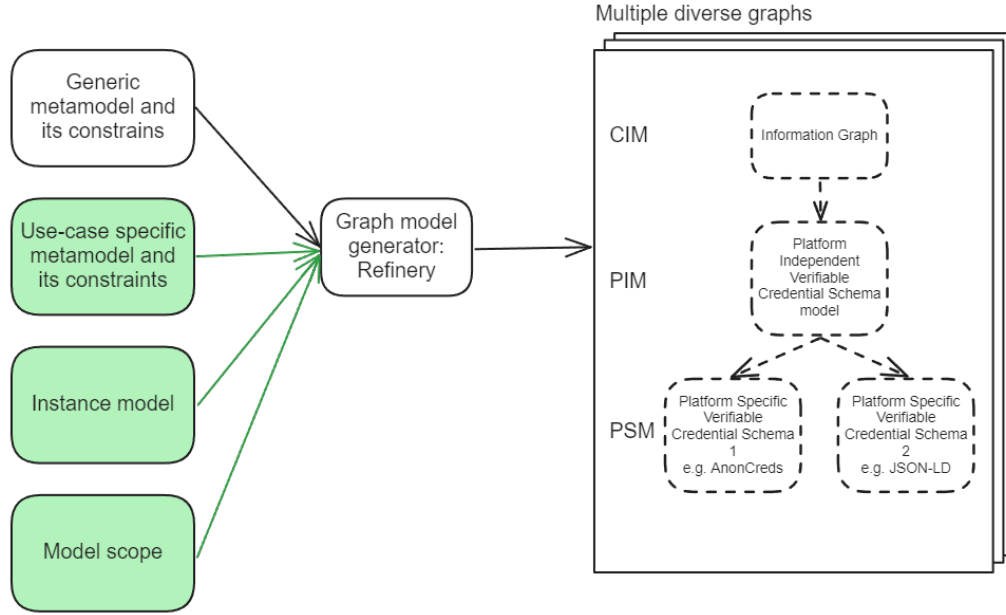


Figure 4.1: The architecture of model generation using Refinery

model is generated, which denotes how are the claims divided between credentials. Finally, each abstract credential is transformed to the carrying format specific representation.

Because these abstraction layers follow the MDA approach, designers can reap the benefits mentioned in Section 2.1.5. Here, this specifically means that the elements of the model are traceable between abstraction layers - which claim from the KG is in which VCS - , and the *transformations* are both clearly defined and extensible between abstraction layers.

4.1.1.1 CIM: Knowledge graph

The first part of the metamodel models the KGs containing the claims the designers want to represent in the Verifiable Credential Schema Ecosystem (VCSE). It follows the subject-property-value relationship model. An example model can be seen on Figure 4.2. This is the layer designers may refine the model with the business level characteristics of the claims - e.g. what level of risk does the unwanted exposure of a claim poses.

4.1.1.2 PIM: Abstract VC Schemas

The second abstraction layer is the abstract model of the VCSs, following the W3C Data Model [1] (See Section 2.1.1.2). The relationships are “partitioned” into claims

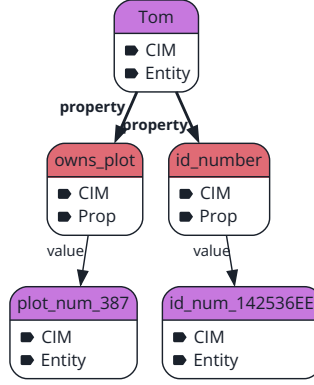


Figure 4.2: An example Knowledge graph

within VCs in a traceable way, meaning there is a one-to-one correspondence between the relationships in the KG and the claims they turned into within the credentials schemas. This forms the abstract VCSE. For an example, see Figure 4.3.

The transformation that takes a KG to an abstract VCS model is systematic separation of entities in the KG into multiple entities in the VCS model so that they fall into multiple credentials, and also satisfy the graph-based requirements the designer extends the meta-model with. This means the model can be extended to satisfy complex requirements that can be stated as patterns on an KG. Examples of such requirements include:

- Two specific relationships must go into the same or separate credentials, based on some rule
- A credential must contain at most or at least two claims
- High-risk and low-risk credentials must never go in the same credential

Here we must recognize that the reconstruction of the KG from the credentials happens using *entity alignment*. Accepting the truthfulness of an alignment - meaning that an alignment presented to the Verifier was intended by the issuers, and not just a coincidence, which the Holder tries to exploit (On Figure 4.3: *a_Tom1* and *a_Tom2* *really* refer to the same Tom) is left to the judgement of the Verifier, similarly to the overall *truthfulness* of the claims, as I mentioned in Section 2.1.1.2.

Essentially, the KG is “split” into disjoint subgraphs where *entity alignment* takes place. Each continuous subgraph is a credential, for which we can decide which is the *credentialSubject* claim, i.e. the “entry point” of the credential.

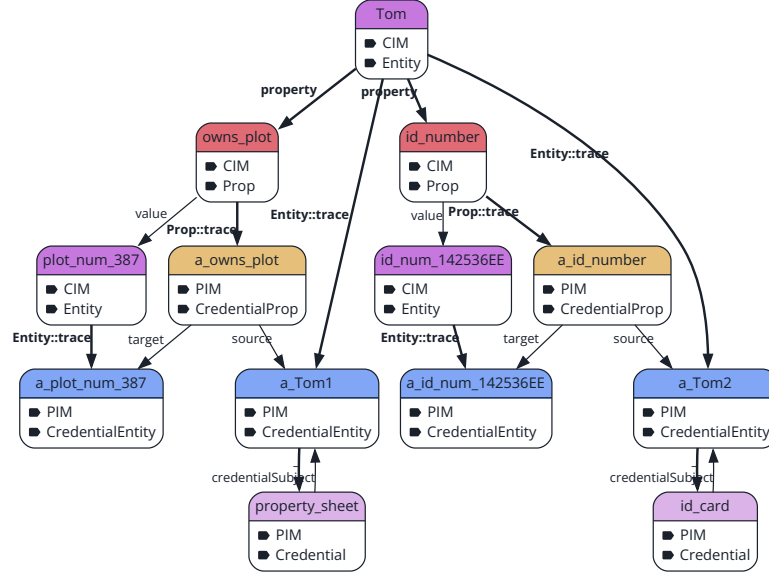


Figure 4.3: An example CIM with PIM

4.1.1.3 PSM: format-specific VC Schemas

Finally, the abstract set of VCSs is refined into carrying format specific representations from which they can be directly implemented. This ensures interoperability across the multiple VC carrying formats, in the sense that the starting KG can be reconstructed from claims that are contained in credentials with different formats, but they are part of the same abstract VCSE model.

As I mentioned in Section 2.1.1.2, these formats offer different characteristics, mainly in the context of what kind of privacy-preserving techniques they enable. Designers can not only enable interoperability between Issuers offering different carrying formats, but also can define technical-level, potentially format specific requirements.

This is the final layer of abstraction, where the technical level requirements can be defined - e.g. credentials containing sensitive information should use a privacy-enhanced carrying format, or credentials that are to be verified by IoT devices should use an easy-to-verify format.

4.2 Partial graph meta-model

The generic meta-model (Figure 4.5) is implemented as an extensible Partial Graph Model. It is easily extended with specific graph-based requirements and concrete model use-case specific concrete model elements. In this section, I enumerate the elements and constraints of the meta-model, and explain how do they fit into its

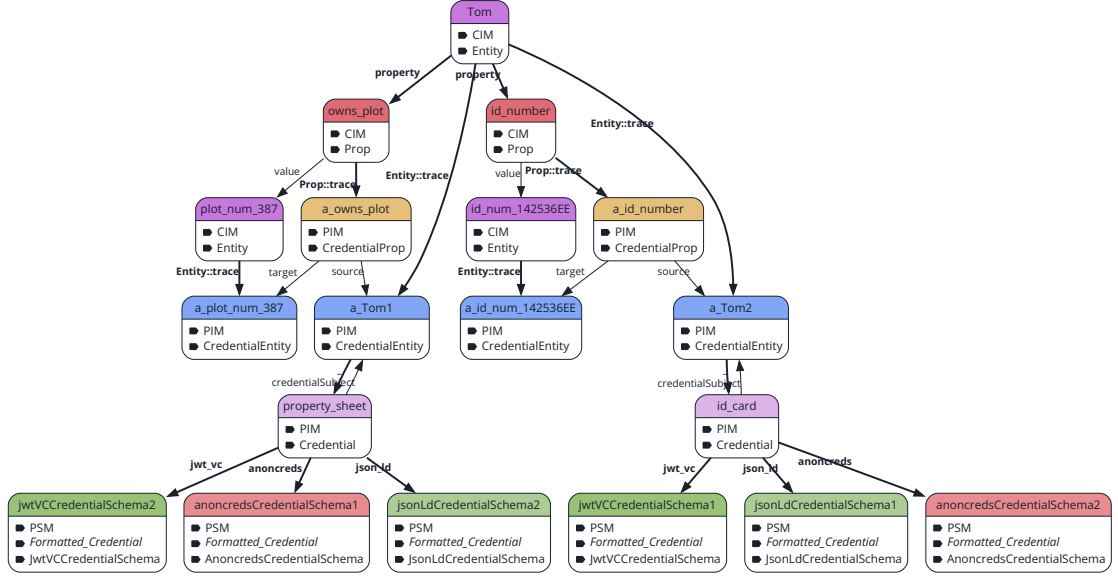


Figure 4.4: Example complete model

larger context. With this metamodel, I was able to generate complete models along with the previously defined architecture. The description of the predicates which define the model can be found in Appendix A.1.

4.3 Evaluating the generated models for DSE

As introduced in Section 2.1.4 and argued for in Section 3.1.5, established DSE techniques offer a wide range of benefits for the designers of VCSEs. In this specific approach, my aim was to first establish the feasibility of such techniques, and explore how they are amenable to specific use-cases. In this section, I detail some rudimentary evaluations which can be done on these simple models that I used to demonstrate the approach.

Expert driven One very useful way that these models can be analyzed is by looking at them with the aim of improving them with new heuristics and identifying implicit constraints and requirements. Sometimes we are looking for unexpected solutions, which can be found by looking at a diverse set of solutions of the *current* iteration. This method offers a key benefit beyond using a solver and unchanging systematic evaluation techniques, as it allows for the iterative discovery of useful implicit constraints. When something doesn't look as expected in the solution, that is an opportunity to learn something new.

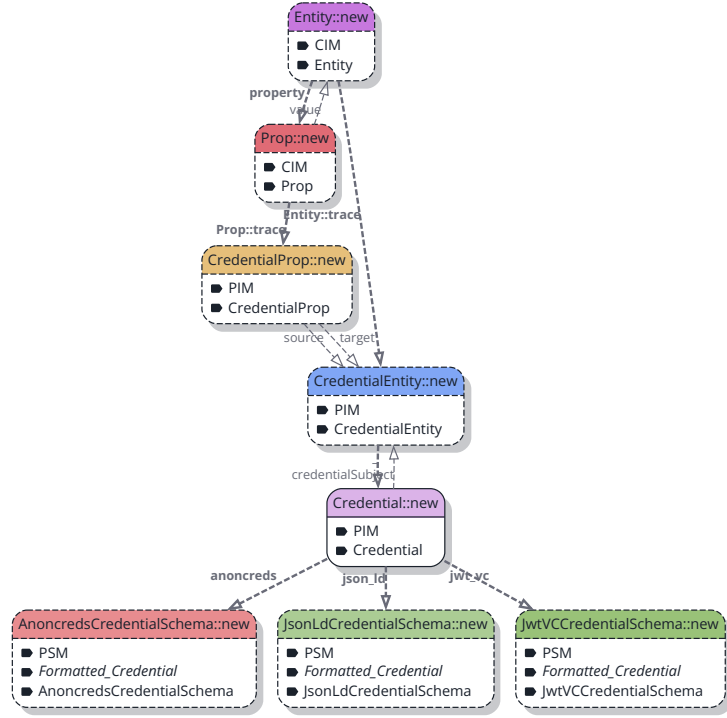


Figure 4.5: Initial partial model in Refinery

Ranking based on a metric Another approach which is possible with these models is to define a metric which can be seen as useful, and rank the generated models based on that. A trivial example could be to count the number of individual Verifiable Credential Schemas that are generated from the Knowledge Graph. With this metric, we can identify the model in which the schemas are the coarsest. This can be useful in a context where the publication of schemas or the issuance of credentials costs a large amount of some expensive resource (*e.g. gas on Ethereum*), or in an iteration step where we want to explore the design where schemas are divided as little as possible.

Pareto Analysis of multi-objective functions Multi-objective optimization of the models is useful when we can identify multiple metrics which are beneficial to optimize, and we cannot combine them into a single metric. An example could include the *Pareto front* formed in the context where we want to evaluate the granularity and the posed correlation risk. This can inform us which models are *Pareto optimal*, meaning the credentials cannot be made less granular without increasing the risk of correlation, and vice versa.

4.4 Assessment of the current implementation

With the current implementation of my approach², I was able to generate consistent VCS models, which satisfied graph-based requirements.

It is in development and currently doesn't implement the full design flow. Right now, I am working on the meta-model, making it more efficient and easy to use. I want to create a rich set of examples, big and small, over which the model generation can be evaluated, with respect to requirement satisfaction, ease of use. and generation performance.

My aim is to also define and create generic DSE evaluation processes, both to showcase the capabilities of my approach, and to provide templates for designers that they can use in their domain specific design endeavors.

The final goal is to provide a toolchain in which designers can iteratively define domain-specific partial models, generate a diverse set of complete models, and evaluate them based on their own metrics.

²Assets are available on GitHub at <https://github.com/BlackLight54/dse-vc-refinery>

Chapter 5

Conclusion and Future work

In this paper, I proposed a novel method for the systematic design of Verifiable Credential Schema Ecosystems from Information Graphs through Design Space Exploration. I presented my approach that combines Partial Graph Modeling with Model Driven Architecture principles to generate and evaluate diverse credential schema designs that satisfy complex requirements while maintaining technical interoperability.

The proposed method addresses a critical gap in the rapidly developing field of digital identity management - the need for systematic, algorithmic approaches to designing interrelated sets of Verifiable Credential Schemas that satisfy complex requirements while maintaining privacy and security characteristics. My contributions include:

- A systematic method for generating diverse credential schema designs using Partial Graph Modeling
- A Model Driven Architecture-based approach ensuring interoperability between different VC formats
- An iterative framework for evaluating generated designs against diverse requirements
- A proof-of-concept implementation demonstrating the feasibility of automated credential schema generation

The initial results demonstrate that systematic modeling and algorithmic generation of credential schemas is not only possible but can provide valuable insights into design trade-offs that might be missed through manual design approaches. The Model Driven Architecture-based approach ensures that generated designs maintain consistency across different levels of abstraction with the ability to provide format-specific representations.

However, there are several limitations in the current implementation that need to be addressed. Future work includes expanding the starting meta-model, optimizing the performance of the generation for large-scale models, and identifying commonalities between domain-specific models that would provide a benefit if they are integrated into meta-model. Also, metrics and benchmarks need to be implemented, with the aim of giving the designers the ability to evaluate the quality of the models along generic measures, while also giving them an example they can follow and extend. These metrics need to be integrated into an iterative, partially automated evaluation process. Furthermore, the approach must be evaluated on actual industry use-cases, be it with existing or emerging Verifiable Credential ecosystems. Finally, the model must be extended with format-specific transformations for all major VC formats.

The proposed approach lays a foundation for more rigorous and systematic development of Verifiable Credential Schema, particularly in complex domains where manual design becomes intractable. While further work is needed to fully realize its potential, the methodology provides a promising direction for addressing the growing challenges in digital identity management and credential system design.

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Appendix

A.1 Explanation of model elements

KG entities

Entity: Represents a subject or a value in the subject-property-value scheme

Prop: Represents a property in the subject-property-value scheme.

KG predicates and rules

statement(Entity, Prop ,Entity): Defines the subject-property-value relationships in the model

neighbours(Entity, Entity): Two entities are neighbours if they are part of a subject-property-value relationship

error non_connected(Entity, Entity): If there are two entities two are not transitively neighbours, meaning there is no path between them in the knowledge graph, the the graph is non continuous, which I disallow, as I want to model *interrelated* VCSEs

PIM entities

CredEntity: Represents a subject or a value in the abstract VC schema model. It is associated with a **Entity**, and multiple **CredEntities** can be associated with one **Entity** - which means they are *aligned*

Claim: Represents a property or a value in the abstract VC schema model. It is associated with a **Prop**, one-to-one, as each property becomes one claim. Its *source* attribute points to the subject **CredEntity**, its *target* attribute points to the value **CredEntity**

Credential: Represents the “entry point” of the Verifiable Credential. It is the root of the continuous sub-graph defined by the **CredEntities** and **Claims**.

PIM predicates and rules

credential_statement(CredEntity, Claim, CredEntity): Defines the subject-property-value relationships of claims within the abstract VC schema model

propagation rule **pim_and_cim_should_align**(Entity, Prop, Entity): If a **credential_statement** is defined, they there should be a corresponding **statement**.

propagation rule **tracing_cim_to_pim...**(...): If there is a **credential_statement** and a **statement** where the **Claim** and **Prop** are associated, then the corresponding entities should be associated too.

aligned(CredEntity, CredEntity): The two **CredEntities** are associated with the same **Entity**m, meaning they are *aligned*.

propagation rule **prop_source**(Claim,CredEntity): The *source* of a **Claim** must be associated with the *subject* of a **Prop**

propagation rule **prop_target**(Claim,CredEntity): The *target* of a **Claim** must be associated with the *value* of a **Prop**

error **no_cred_props_for_cred_entity**(CredEntity): Each **CredEntity** must be part of a subject-property-value relationship.

Root_cred_entity(CredEntity): **CredEntity** that is the root of its subgraph

error **root_ent_doesnt_have_cred**(Root_cred_entity): Each root must have an associated Credential.

PSM entites

abstract **Formatted_Credential**: Abstract representation of the carrying format specific credential schemas

AnoncredsCredentialSchema; JsonLdCredentialSchema; ...: Specific representation of the carrying format specific credential schemas